SRS-Template

Clinic Reservation System

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1. Project Overview

The Clinic Reservation System stores a list of doctor appointments and allows the patient to schedule an appointment, update the appointment status, and maintain all previously made reservations. Furthermore, the system allows the doctor to access and maintain their relevant appointments and patient information.

2. Project Team

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1. Introduction

A Clinic Reservation System is needed to improve the way clinics manage patient appointments. Manual booking methods often lead to delays, scheduling conflicts, and inefficient use of staff time. The system allows patients to schedule, update, and manage appointments easily, while doctors and administrative staff can view and organize schedules more efficiently. It can also work with external systems such as Electronic Health Records (EHR) to keep patient information accurate and up to date. Overall, the system helps clinics improve efficiency, reduce missed appointments, and provide better healthcare services.

2. User Requirements

This section lists the system requirements described directly from the perspective of the primary end-users, using their own non-technical language.

2.1 Patient Requirements

- I need to be able to securely book, reschedule, or cancel my appointments online 24/7 without having to call the clinic.
- I want to search for doctors based on their medical specialty, location, and real-time schedule availability.
- I want to receive automatic text or email reminders before my visit so I do not forget.
- I need to fill out my medical history forms and upload my insurance documents online before I arrive at the clinic.

2.2 Clinical Provider (Doctor) Requirements

- I need my calendar to update instantly when a patient books online so that my time is never double-booked.
- I want the ability to customize my schedule by blocking out specific times for surgeries, telehealth calls, or routine follow-ups.
- I need the system to connect with patient medical records so I have their full health history ready to review before the appointment begins.

2.3 Administrative Staff Requirements

- I need a central digital dashboard to easily manage schedules for multiple doctors and branch locations at the same time.
- I want the software to automatically handle routine reminder calls and booking confirmations so I can focus on helping the patients who are physically in the waiting room.
- I need the system to automatically offer canceled time slots to other patients on a waitlist so the doctor's time is not wasted.

3. Functional Requirements

1. Functional requirement Name: Appointment Booking

- **Description/ Action:** Patient schedules an appointment by selecting a doctor, date, and time.
- **Requirements/ Inputs:** Selected doctor, desired date, and preferred time slot.
- **Source:** Patient input via the self-service web or mobile portal.
- **Pre-condition:** The selected doctor must have available time slots on the chosen date.
- **Post-condition:** The selected time slot is locked and reserved for the patient.
- **Output:** A booking confirmation message displayed to the user.

2. Functional requirement Name: Appointment Management

- **Description/ Action:** Patient updates or cancels an appointment, and the system updates availability accordingly.
- **Requirements/ Inputs:** Appointment ID, cancellation trigger, or new date/time selection.
- **Source:** Patient input through the system interface.
- **Pre-condition:** An active, future appointment must exist for that patient.
- **Post-condition:** The appointment is updated or canceled, and the time slot is made available again.
- **Output:** An updated appointment status display and an updated master schedule.

3. Functional requirement Name: Doctor Schedule Management

- **Description/ Action:** Admin defines available time slots and modifies working hours.
- **Requirements/ Inputs:** Specific calendar dates, start/end times, and block-scheduling rules.
- **Source:** Clinical provider or admin input via the calendar dashboard.
- **Pre-condition:** The user must be authenticated with provider or administrator access privileges.

- **Post-condition:** The system updates the doctor’s schedule with the new availability.
- **Output:** An updated, real-time provider calendar view.

4. Functional requirement Name: Patient Record Access

- **Description/ Action:** Staff views patient information and links scheduled appointments to patient records.
- **Requirements/ Inputs:** Patient demographic data, intake forms, and appointment metadata.
- **Source:** Electronic Health Record (EHR) database and the patient's digital intake workflow.
- **Pre-condition:** A patient profile must exist and be authenticated within the system.
- **Post-condition:** The appointment automatically updates in both the booking system and the patient's permanent, long-term medical file.
- **Output**: A single, organized screen showing the patient's history and upcoming visits.

5. Functional requirement Name: Notifications

- **Description/ Action:** System automatically sends booking confirmations and reminders to the patient.
- **Requirements/ Inputs:** Confirmed appointment timestamps and patient contact preferences.
- **Source:** The system generates notifications automatically using appointment details and patient contact information
- **Pre-condition:** An appointment must be in a 'Confirmed' state.
- **Post-condition:** The notification is sent to the communication service and prepared for delivery to the patient.
- **Output:** An SMS, email, or push notification delivered to the patient.

6. Functional requirement Name: User Registration

- **Description/ Action:** A new user (e.g., Patient) creates an account to access the clinic reservation system.

- **Requirements/Inputs:** User's full name, email address, phone number, and a secure password.
- **Source:** User input via the web or mobile portal.
- **Pre-condition:** The user does not already have an active account associated with the provided email address.
- **Post-condition:** A new user profile is created, and the credentials are securely stored in the system database.
- **Output:** A successful registration message displayed to the user and redirection to the login screen.

4. Non-Functional Requirements

- **Security and Privacy:**
  The system shall protect patient data and ensure privacy by complying with standards such as HIPAA and GDPR.
- **Reliability and Availability:**
  The system shall be reliable and remain available with minimal downtime during normal operation.
- **Usability and Accessibility:**
  The system shall provide a simple and user-friendly interface accessible on both web and mobile devices.
- **Performance and Scalability:**
  The system shall respond quickly and handle multiple users efficiently without performance degradation.
- **Compatibility and Integration:**
  System Connection and Sharing: The system must easily connect and share data with external software, like Electronic Health Records (EHR—the digital version of a patient's medical charts)